Fixed-charge selection, stability and bounded model comparison

TECT verification-first repository
claim `C6-SPACETIME-SIGNATURE`

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1 Purpose and scope

P1, PAH and TECT-EXCH-001 are preserved byte for byte. This checkpoint changes the comparison, not the parent. EXCH002 fixes the single charge

$$\bar{Q} = 11008/27 = V\rho_*/2, \quad V = 4096, \quad \rho_* = 43/216, \quad q = \bar{Q}/V = 43/432.$$

This is an explicit research input inherited from R158, not an observation. At each fixed finite $K \geq 1$, retain the exact side-16 torus, Fourier cutoffs K for $\Psi \in \mathbb{C}^3$ and $2K$ for $\chi \in \mathbb{C}$, orthogonal projections, real L^2 pairing and all P1 positive floors. The unchanged parent and charge are

$$H = F_{\mathrm{P1}} + \int \left(\frac{|\chi|^2}{2} + \frac{|\chi|^4}{4} - \epsilon \Re(\bar{\chi} S) \right), \quad S = \Psi^T \Psi, \quad \epsilon = 1/16, \qquad Q = \frac{1}{2}\|\Psi\|_2^2 + \|\chi\|_2^2.$$

The bracket is $\{z, \bar{z}\} = -2i$ in orthonormal coordinates, with the same canonical parameter τ as EXCH001. No Gibbs state, actual clock, reservoir reset or new dissipation is inserted. Let $\Sigma = \{Q = \bar{Q}\}$. All norms and distances below are finite-coordinate L^2 norms at fixed K.

The preregistered contracts are EXCH002-v1 and ALT001-v1, SHA-256 respectively

```
5dad4605d24d338d0ff4a8c169607595d7c59f247c15a4388185fceee31d75cc
75e8a1e70a27a9880bdcc17a74f812dbeb95a2be4f6cb4ef9f51c98ff3cfe060
```

The alternative models are separate definitions, not repairs of this parent.

2 Selection on one nonzero sector

For every reservoir-only state on Σ, Cauchy–Schwarz gives

$$H(0, \chi) \geq E_R = V(q/2 + q^2/4), \qquad E_R/V = 39001/746496.$$

A constant-modulus reservoir attains it. R158 on its unchanged component gives the best pure-order energy $E_P = \mu_t \bar{Q}$, attained by the lowest internal real eigenvector v_* and shell $k = (2\pi/16)(1,1,1)$. Since $1/4 < \mu_t < 1/3$, $E_P/V < 43/1296 < E_R/V$.

Now use only the preregistered same-charge trial

$$r = \epsilon\rho_* = 43/3456, \quad p = \rho_* - 2r^2 > 0, \qquad \Psi = \sqrt{p}\, v_* e^{ikx}, \quad \chi = re^{2ikx}.$$

Its charge is exactly $V(p/2 + r^2) = \bar{Q}$ and its doubled mode is admitted. The arbitrary-amplitude plane family has vanishing Class-II currents; we do not import R158's fixed-charge minimization theorem at $p < \rho_*$. Using its exact plane energy instead,

$$\frac{H_{\mathrm{tr}} - E_P}{V} = (1/2 - \mu_t) r^2 + \frac{r^4}{4} + \frac{2\gamma}{3} p r^4 - \epsilon r p < 0, \qquad \gamma = 81/50.$$

Replacing μ_t by $1/4$ gives a strictly negative exact rational upper bound, less than $-1/10000$. Thus $\min_\Sigma H < E_P < E_R$. The finite-dimensional ellipsoid Σ is compact and H is smooth. Its global minimum set $\mathcal{M} = \arg\min_\Sigma H$ is nonempty and compact, and *every* member has both $\Psi \neq 0$ and $\chi \neq 0$. No morphology, uniqueness or individual orbit is selected by this argument.

3 Variational saddle versus actual reservoir dynamics

The declared constant reservoir has the exact relative orbit

$$R(\tau) = (0, \sqrt{q}\, e^{-i(1+q)\tau}), \qquad \mu_R = (1+q)/2 = 475/864.$$

On the same-charge curve $\Psi = tv_* e^{ikx}$, $\chi = \sqrt{q - t^2/2}$ spatially constant, the interaction integrates to zero. The energy change per volume is

$$\tfrac{1}{2}(\lambda_0 - \mu_R)t^2 + (\lambda/4 + 1/16)t^4 + \gamma t^6/6.$$

Its quadratic coefficient is negative. A positive constrained direction is provided by $\Psi = 0$ and $\chi = \sqrt{q}(1 + t\cos(2\pi x_1/16))/\sqrt{1 + t^2/2}$: its quartic average exceeds q^2 by $q^2(2t^2 + t^4/8)/(1 + t^2/2)^2 > 0$. Hence the reference is a constrained energetic saddle.

The actual growing mode is not the downhill shell. At $\Psi = 0$, positive-floor Class-II currents are at least quadratic in Ψ, so their energy is of order four and does not change the linearized L block. In the co-rotating frame generated by $H - \mu_R Q$, the linearized order equation is $i\dot{\psi} = (L - \mu_R)\psi - 2\epsilon\sqrt{q}\,\bar{\psi}$. For a real internal eigenbranch and Fourier pair $n, -n$, the Bogoliubov matrix is

$$i\partial_\tau \begin{pmatrix} a_n \\ \bar{a}_{-n} \end{pmatrix} = \begin{pmatrix} d & -c \\ c & -d \end{pmatrix} \begin{pmatrix} a_n \\ \bar{a}_{-n} \end{pmatrix}, \quad d = \lambda_j(n) - \mu_R, \quad c = 2\epsilon\sqrt{q}.$$

The time generator squares to $(c^2 - d^2)I$. The real internal matrix is derived from original masses, lock and z_0: $M = \mathrm{diag}(m_a) + k_{\mathrm{lock}}(I - z_0 z_0^T/|z_0|^2)$. Exact root counts isolate its largest eigenvalue in $(1/5, 21/100)$. Alternating-series Machin bounds certify $3 < \pi < 22/7$; monotonicity of the pinned quadratic spatial symbol then gives, for $n = (1, 0, 0)$,

$$11/20 < \lambda_{\mathrm{high}}(n) < 29/50, \qquad c^2 - d^2 > \frac{43}{27648} - \left(\frac{29}{50} - \frac{475}{864}\right)^2 = \frac{18701}{29160000} > 0.$$

This mode exists for every $K \geq 1$, is tangent to Σ at R and is transverse to its phase orbit. It proves actual exponential linear growth in the original canonical parameter. In contrast, the lowest R158 shell has $c^2 - (\lambda_0 - \mu_R)^2 < 0$: that downhill block is elliptic. Energetic instability alone was not used as a dynamical instability theorem.

Nonlinear orbital instability of this one reference. Teschl, *Ordinary Differential Equations and Dynamical Systems*, author's AMS preliminary version, section 9.2, Theorem 9.4, printed p.259 (PDF p.270), provides a local unstable manifold even with center directions. The official PDF was hash-pinned and the complete relevant page inspected: https://www.mat.univie.ac.at/~gerald/ftp/book-ode/ode.pdf. Its SHA-256 is

`362433156525216abf596c17ce843204510e96d57afa4284a37c7aa5a9ffc36e`

Crosswalk: fixed finite dimension, smooth vector field, co-rotating fixed point and positive transverse eigenvalue all hold. No hyperbolic-only Hartman–Grobman theorem is imported. A point on the local unstable manifold approaches $R(0)$ backward in time. Conservation/continuity force its charge to equal $\bar{Q}$. Choose such a point with $\Psi \neq 0$, possible by tangency to the certified positive eigenspace. Its backward iterates approach the reservoir orbit, but evolve forward to a phase-rotated point with fixed nonzero $\|\Psi\|_2$, bounded away from that orbit. This proves nonlinear orbital instability within the fixed sector. It does not classify all reservoir-only phases. The exact $\Psi = 0$ orbit itself remains invariant.

4 Stable minimum set is not an attractor

EXCH001 already supplies complete finite-K flow conserving H and Q. Consequently $\mathcal{M}$ is invariant. For any distance $\delta > 0$, the compact sector subset at distance at least δ from $\mathcal{M}$ has a positive energy gap above $m = \min_\Sigma H$ (if the subset is empty, there is nothing to prove). Uniform continuity supplies a smaller initial neighborhood with energy below that gap. Energy conservation prevents escape. This proves two-sided set-Lyapunov stability, not stability of every individual orbit, a unique condensate or cutoff-uniform bounds.

For any initial state outside $\mathcal{M}$, $H > m$ stays constant. If its distance to $\mathcal{M}$ tended to zero, compactness and continuity would force its energy to tend to m, a contradiction. Thus no outside *full-state* trajectory converges to the ground set. This does not exclude coarse-observable relaxation or transient structures. The unstable reservoir manifold has energy $E_R > m$, not the mixed minimum energy.

5 Separate bounded alternative intake

ALT-DENSITY replaces pair conversion by $\eta|\chi|^2|\Psi|^2$ in a separate model. Its two separate phase symmetries conserve both component charges, so it fails the required transfer filter. No further sweep is run.

ALT-TENSOR instead uses $C \in \mathrm{Sym}_3(\mathbb{C})$ with six Frobenius-orthonormal coordinates of weight two, the same cutoff, bracket, charge, coefficients and sector. Its interaction is $-\epsilon\Re\,\mathrm{tr}(C^\dagger\Psi\Psi^T)$ and reservoir energy is $\|C\|_F^2/2 + \|C\|_F^4/4$. Off-diagonal basis matrices are divided by $\sqrt{2}$; duplicate counting is forbidden. For $P = \Psi\Psi^T$,

$$\|P\|_F^2 = \rho^2, \quad p = (\psi_1^2, \psi_2^2, \psi_3^2, \sqrt{2}\psi_1\psi_2, \sqrt{2}\psi_1\psi_3, \sqrt{2}\psi_2\psi_3).$$

The weighted transfer cancellation is unchanged. Constant unitary $\Psi \mapsto U\Psi$, $C \mapsto UCU^T$ preserves the added interaction and reservoir rule. This is *not* a symmetry assertion about full P1.

At zero reservoir, both first charge derivatives vanish. The full-law second derivatives are

$$Q_C''(0) = \epsilon^2 \int \rho^2, \qquad Q_\chi''(0) = \epsilon^2 \int |\Psi^T\Psi|^2.$$

P1 and pure-reservoir contributions vanish in these derivatives at zero reservoir; this is not an interaction-only approximation. The same fixed-charge state $\Psi = \sqrt{q}(1, i, 0)$, reservoir zero, has identical initial P1 energy in both models, but gives respectively $1849/2916 > 0$ and zero. Thus the tensor removes instantaneous internal pair-source cancellation. Scalar cancellation need not persist under P1. No all-time transfer dominance, faster relaxation or stability superiority follows. Six rather than one complex reservoir coordinates is a real cost. Rank-one real aligned fields reproduce scalar interaction/charge, but a full invariant dynamical embedding needs P1 span preservation. In particular $C = \chi I$ is not charge-preserving. Overall superiority remains unestablished.

6 Devil's-advocate and verification

1. **A trial comparison proves global selection (VALID, controlled).** The trial beats the rigorous minima of both pure classes, and compactness supplies a full constrained minimum; no trial morphology is declared unique.

2. **Negative Hessian proves growing time modes (UPHELD boundary).** The actual rotating generator was derived separately. The downhill shell is elliptic; the exponential mode uses another spectral branch.

3. **Center spectrum invalidates nonlinear instability (DISMISSED).** Theorem 9.4, not a hyperbolic-only theorem, supplies the needed manifold; charge conservation puts it in the same exact sector.

4. **Stable minima attract the unstable reservoir (UPHELD against it).** Different conserved energies forbid that full-state conclusion.

5. **Tensor is globally better (UPHELD against it).** Only a structural/initial-response advantage is proved, with more degrees of freedom and no full-P1 internal symmetry or stability comparison.

6. **Finite stability supplies physical time or a limit (UPHELD).** Every assertion is at fixed finite K and the original hypothetical canonical parameter. No actual carrier, thermodynamic limit or Pre-A claim.

Primary exact checks, a non-importing independent checker and a separate hostile source/code review support this result. The alternative checker is independently derived. Partial Lean lemmas encode algebra only, not compactness, manifold existence, nonlinear stability or physical correspondence. External adversarial review of the fixed-sector and unstable-manifold crosswalk is invited. Old deterministic-PDF rebuild verifiers were not rerun.

7 Result footer

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Result ID: R-579 (C6-SPACETIME-SIGNATURE)
Precise statement: fixed-Q mixed minimum set is stable;
constant reservoir relative orbit is unstable;
outside full-state attraction to minimum set is impossible.
Scope: Q=11008/27, each fixed finite K, unchanged EXCH001.
Dependencies: pinned P1/R158/R578; Teschl Theorem 9.4.
Evidence grade: ANALYTIC, EXACT, EXECUTED, CONDITIONAL.
Reproduction command: python -X utf8
verification/scripts/exch002_verify.py
Expected output: PASS_SCOPED_PACKAGE and partial Lean.
Falsification gate: wrong sector, source, Hessian or flow.
Tier before / after: C6 T1 / T1; T054 unchanged.
No-overclaim statement: no individual morphology,
physical clock, relaxation, PreA, QFT, gravity or limit.
Next required action: a physical state/observable packet
or precise error; no automatic model repair or successor.
```